Machine Learning or Deep Learning?

A 2-page decision guide for e-commerce product teams. By Kirti Mittal, November 2026.

Product teams often hear "let's use AI" before anyone has asked which kind. Traditional **machine learning (ML)** learns from features that people choose. **Deep learning (DL)** uses neural networks with many layers that learn the features themselves from raw data such as pixels or text. DL is not simply "better ML." It needs far more data and computing power, and it is harder to explain. This guide helps a team pick the right one before it spends a sprint on the wrong one.

FIVE QUESTIONS TO ASK FIRST

1. **Can we name the features ourselves?** If a person can list what matters (order total, number of past returns, words in a subject line), start with ML. If the signal is in raw pixels, audio, or the meaning of sentences, DL is usually needed.
2. **How much labeled data do we have?** ML can work with hundreds or thousands of examples. DL usually needs very large labeled sets, or a pre-trained model you can adapt.
3. **Do we have to explain each decision?** If support or legal teams must say why a customer was flagged or an email was filtered, ML is easier to explain.
4. **What will it cost to run?** ML runs on an ordinary server. DL often needs GPUs to train and more expensive servers to answer quickly.
5. **What happens when it is wrong?** If a mistake is cheap and easy to undo, a simpler model is fine. If a wrong result loses a sale or a customer's trust, test the more accurate option against it before you launch.

QUICK MAP FOR COMMON E-COMMERCE FEATURES

Feature	Data it uses	Start with	Why
Spam or fake-review filter	Words, links, sender, account age	**ML**	Features are easy to name; decisions must be explainable
Customer churn prediction	Order history, contract length, monthly spend	**ML**	Structured table data; works with modest data
Price or demand estimate	Sales history, season, promotions	**ML**	Numeric features; easy to retrain and check
Search by photo / auto-tagging product images	Raw images	**DL**	Visual features are too complex to describe by hand
Translating product descriptions	Text in sequence	**DL**	Meaning depends on context across the whole sentence

TWO REAL-WORLD EXAMPLES

ML fits: spam filtering (Apache SpamAssassin)

SpamAssassin scores each email with hand-written tests and a Bayesian classifier that learns which words show up in spam and non-spam. Its filter starts scoring once it has learned from **200 spam and 200 non-spam** messages (Apache SpamAssassin Project, 2019).

- Features are easy to name
- Little data and an ordinary server
- An admin can see why a message was flagged

Why not DL: more data, more hardware, and less explainable, for a problem simple counting already handles well.

DL fits: visual search (Amazon StyleSnap)

Shoppers upload a photo of an outfit and the app finds similar items. Amazon says it uses computer vision and deep learning to find clothing in a photo and sort it into categories like "fit-and-flare dresses" (Amazon, 2019), built on convolutional neural networks (Amazon Science, 2020).

- Input is raw pixels
- Real photos are messy: angles, lighting, backgrounds
- A large catalog provides the training data

Why not ML: hand-built features like color counts cannot tell a blazer from a cardigan of the same color.

MISTAKES I WOULD AVOID

- **Starting with DL because it sounds more advanced.** Build the simple ML version first. It gives you a baseline that any DL model must beat.
- **Ignoring explainability until a customer complains.** Decide up front who needs to understand a decision and how.
- **Training on biased history.** If past data reflects stock-outs or old rules, both ML and DL will repeat them.
- **Skipping a test against real users.** Compare the model's results with what customers actually do before a full launch.

ONE RULE OF THUMB

"Can I describe the features myself?" If yes, start with machine learning. If not, deep learning is probably the right tool, as long as you have the data and budget it needs.

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